

Francisco Rau

React & TypeScript Engineer — ships production apps solo, API-first, AI-native

+31 6 38859519 | franciscorau08@gmail.com | franciscorau.com | linkedin.com/in/francisco-rau | Tilburg, NL

Summary

Product engineer who builds and ships alone. Founder of Citania, a production SaaS for salons in Mexico: ~80,000 lines of TypeScript across a React web app, an Express/PostgreSQL API and a React Native (Expo) mobile app, with ~294 automated tests and a contract-first OpenAPI setup that generates shared Zod schemas and typed clients so web, mobile and server can never drift. Before that, a year building a Dutch contact centre's internal tooling as React/TypeScript web apps on live APIs. Agentic AI is my default way of working, not an experiment — economics and data-science degrees underneath (Tilburg).

Skills

TypeScript · React · React Native (Expo)
Node/Express · PostgreSQL · REST & OpenAPI (contract-first)
Testing: Vitest, supertest, Testing Library (~294 tests in production)
Integrations: Stripe Connect, Clerk, Google Calendar & Microsoft Graph, Meta WhatsApp API
Agentic AI development (daily practice, production output)
Git · full lifecycle: architecture → build → test → deploy
English (fluent) · Spanish (native) · Dutch (A2)

Experience

Data & Process Analyst — Netvice

May 2025 – Present · Tilburg, Netherlands

- Introduced agentic AI development (Replit, Claude Code) as a delivery method — an early bet that let one analyst design, build, and ship production tooling end to end.
- Real-time operations layer: live dashboards wired to telephony webhooks (agent line state, idle detection, coverage views), plus a Daily Agent Planner, agent day reports, and a finance board (pilot).
- Built everything configurable rather than hard-coded, so the tooling keeps working as campaigns and staff change.
- Fully self-directed: each of the systems below began as my own proposal — need identified, pitched, built, and shipped at pace.
- Grew from data intern to Data & Process Analyst, working with increasingly senior stakeholders — ultimately on strategic projects directly with the company's founder.
- Established operational BI as a new, self-owned workstream — a first in the company's ten-year history — with full ownership from roadmap to daily use on the call-center floor.
- Forecasting model (the centerpiece): turns planned mailing volume into the exact number of agents needed on any future day — a cohort-based simulation of call attempts, pickup decay, and callback flows. Because it runs off the mailing calendar, agent demand is known 500+ days ahead, far beyond the notice period scheduling requires. Includes capacity-choke detection and a batch-splitting optimizer.
- Performance analytics: a percentile-based hourly pacing metric that makes agents comparable across campaigns with very different call lengths and success rates, combined with outcomes into a single performance index powering real-time TV rankings on the floor.

Student Researcher — Philips

Sep 2022 – Dec 2022 · Eindhoven, Netherlands

- Researched OneBlade market penetration among students; built linear models estimating adoption of a blade-subscription service across student segments.

Data Entry — Von der Heide

Jan 2019 – Aug 2019 · Mexico City, Mexico

- Cleaned and migrated the company database to a new CRM system, safeguarding data integrity throughout the transfer.

Selected Projects

Market-Gap Analysis: Online-Booking Adoption in Mexico City 2021 · Google Places API, headless browser, Streamlit

- End-to-end data pipeline measuring online-booking adoption across 10,605 businesses: 722 structured API queries, deduplication, two independent booking detectors, and public-registry enrichment. Measured 6.9% adoption; delivered a 9,878-row qualified dataset (89% with direct contact details) as a workbook and dashboard.

Weather Prediction-Market Research 2026 · Python, probability calibration, market microstructure

- Tested whether prediction markets price official weather observations efficiently: implied-probability extraction from order books, no-arbitrage coherence checks, repricing-latency measurement. Documented an honest null result.

Citania — Booking & Payments Platform 2025–2026 · TypeScript, PostgreSQL, React Native

- Solo-built multi-tenant booking and payments SaaS: 100+ integration tests, database-level slot exclusivity, two-way Google Calendar sync. Live in production.

Football Player Value Modelling 2024 · Python, statistical modelling

- Predicted market values on a multi-year, 100+ competition dataset compiled myself. Best model: $R^2 = 0.921$, RMSE \$2.23M.

Education

MSc Data Science & Society (Business Track) — Tilburg University 2023 – 2024

BSc Economics — Tilburg University 2020 – 2023

Languages

English (native), Spanish (native), Dutch (A2, actively learning)